

JKBOSE 2026

CLASS 11TH

Bhaichara
ON
TOP.

#XUVV♥
ORGANIC CHEMISTRY:
SOME BASIC PRINCIPLES
AND TECHNIQUES

LECTURE- 12 **CHEMISTRY**

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Topics To Covered

- 1** **Concept of Electrophile and Nucleophile** (Part - 2)
- 2** **Covalent Bond Fission and Intermediates**

7 DAYS ^(Aug 1 to 7) BOARD CHALLENGE



COMPLETE YOUR BOARD SYLLABUS



NCERT REVISION

Strengthen Concepts.
Build Clarity.



MCQ PRACTICE

Practice More.
Score More.



FORMULA SHEET

Quick Revision.
Smart Recall.

DAY 1

Some basic
Concepts of
Chemistry
(First half) ^(Mans)
Short notes.

DAY 2

Some basic
Concepts of
Chemistry
(Other half)
(Short notes).

DAY 3

Complete
Revision
of
1st chapter
+
PYQ.

DAY 4

STRUCTURE
OF
ATOM
(FIRST HALF)
HEISENBERG.
(Short notes)

DAY 5

STR OF
ATOM
Part-2.
(Short-notes)

DAY 6

Complete
Str. of
Atom
+
PYQ.
(2024-)

DAY 7

Spec +
STR OF
ATOM
(Short
notes)



SET A TARGET
EVERY DAY



COMPLETE
YOUR TARGET



SHARE DONE
IN THE GROUP



STAY CONSISTENT
STAY AHEAD

CONCEPT OF ELECTROPHILE AND NUCLEOPHILE $\rightarrow$



1) ELECTROPHILE $\rightarrow$ Electrophiles are electron loving species. Their attraction for electron is due to the presence of $\bar{e}$ deficient atom in them. Electrophiles may be positively charged or Neutral species.

a) Positively charged electrophiles $\rightarrow H^+, Cl^+, I^+, NO^+, R^+, NO_2^+$ etc.

b) Neutral electrophiles $\rightarrow AlCl_3, BF_3, SnCl_4, FeCl_3, \dot{R}, SO_3$ etc.

• Electrophiles behave as Lewis acids ($\bar{e}$ pair acceptor).

Note $\rightarrow NH_4^+, Na^+, Ca^{+2}$ etc can't act as electrophile because their outermost shell is full.

• E^+

2) NUCLEOPHILE $\rightarrow$ are nucleus (+ve charge) loving species. These are $\bar{e}$ rich species containing at least one lone pair of $\bar{e}$. It can be negatively charged or neutral.

a) Negatively charged $\rightarrow H^-, Cl^-, I^-, Br^-, R^-, OH^-, NO_2^-, CN^-$ etc.

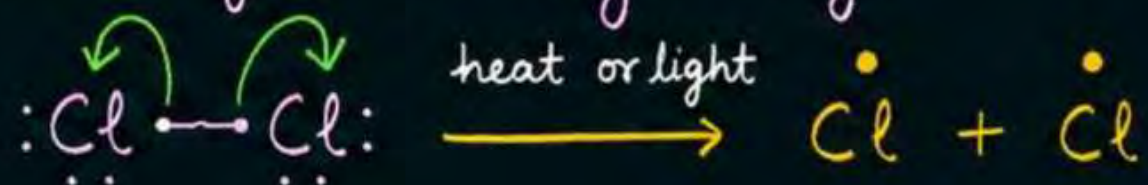
b) Neutral nucleophiles $\rightarrow H_2O:, NH_3:, R-OH, R\ddot{O}H, R-\ddot{N}H_2$ etc.

• Nucleophiles act as Lewis base ($\bar{e}$ pair donor).

• Nu^-

FISSION OF COVALENT BOND.

1) HOMOLYTIC CLEAVAGE $\rightarrow$ If the covalent bond breaks in such a way that each atom takes away one $\bar{e}$ of shared pair (bond), it is called homolytic cleavage. e.g.

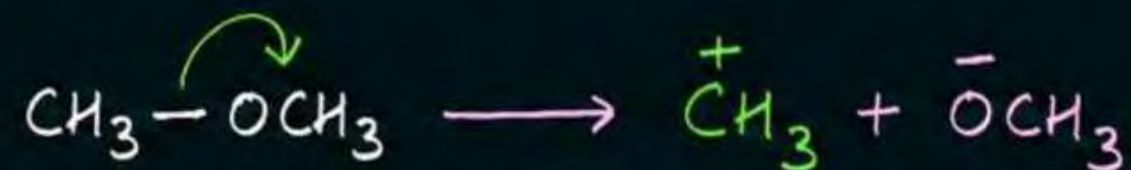


Free Radical

- These odd $\bar{e}$ species are called free Radicals.
- Homolytic cleavage is favoured by high T, UV light or peroxides.

2) HETEROLYTIC CLEAVAGE $\rightarrow$

If the covalent bond breaks in such a way that both the electrons of covalent bonds are taken away by one of the bonded atom (more electronegative atom), this type of cleavage is known as **HETEROLYTIC CLEAVAGE**. e.g.



Q. Mein Complete Intermediates Ko
Repeat Krn.

a) Yes but handwriting

b) Yes but with ppt.

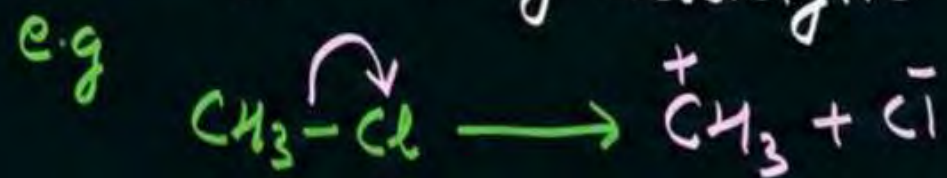
c) No need of repetition

d) Last complete. lecture
repeat.

INTERMEDIATES $\rightarrow$ The short lived highly reactive chemical species through which majority of organic reactions occurs are called reactive intermediates. Some important intermediates are discussed below.

1) CARBOCATION $\rightarrow$ Species having positive charge on carbon and carrying six electrons in their valence shell are called carbocation.

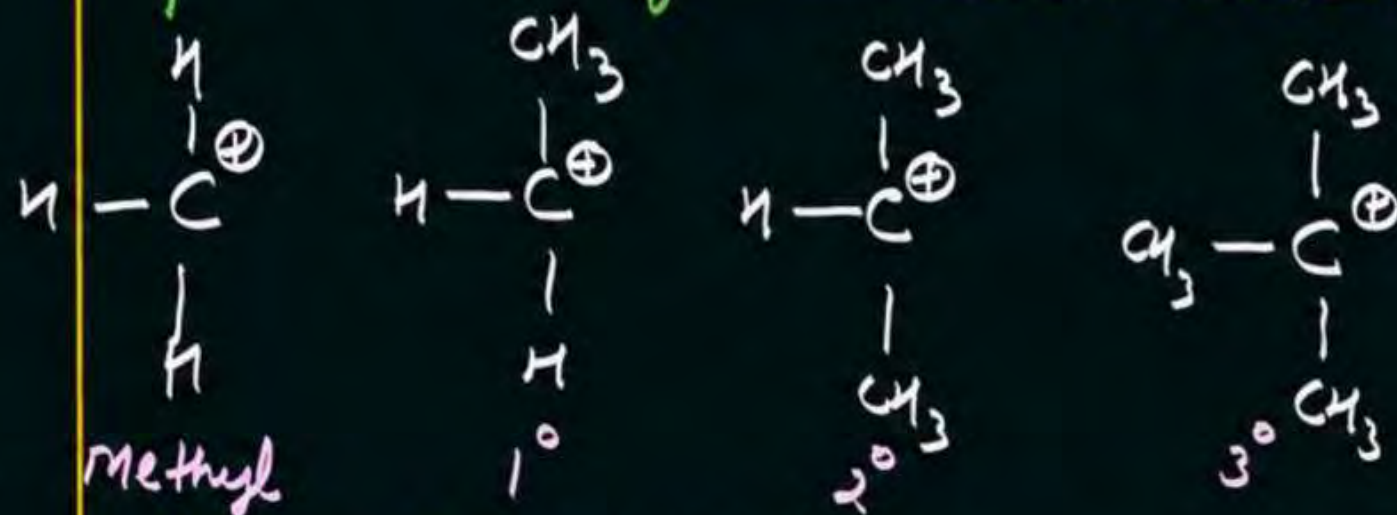
- It is formed by heterolytic cleavage.



- It is $\bar{e}$ deficient species.



Classification $\rightarrow$ Classified as primary (1°), secondary (2°) & Tertiary (3°), depending upon the no. of carbon atoms attached. i.e.



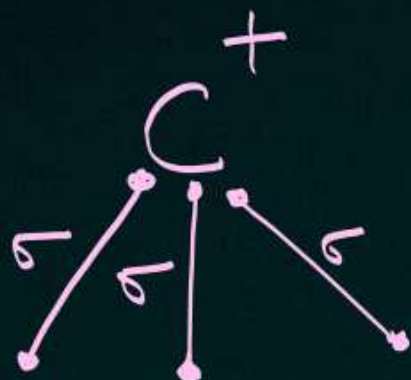
STABILITY $\rightarrow$ Stability follows the order.



because. Carbocation is electron deficient species any $\bar{e}$ donating group stabilise the carbocation.

Tertiary has three $\bar{e}$ donating groups attached (via Inductive & hyperconjugative) i.e. why 3° is most stable.

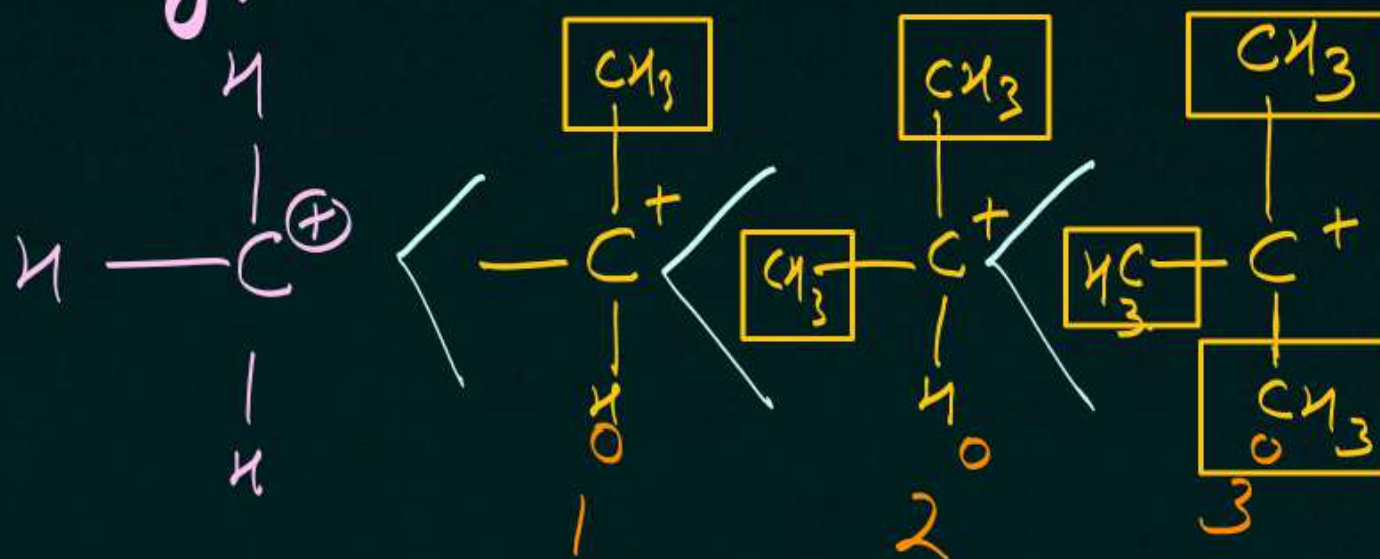
1) Carbocation $\rightarrow$



- $\bar{e}$ deficient species
- Heterolytic cleavage.



Types.

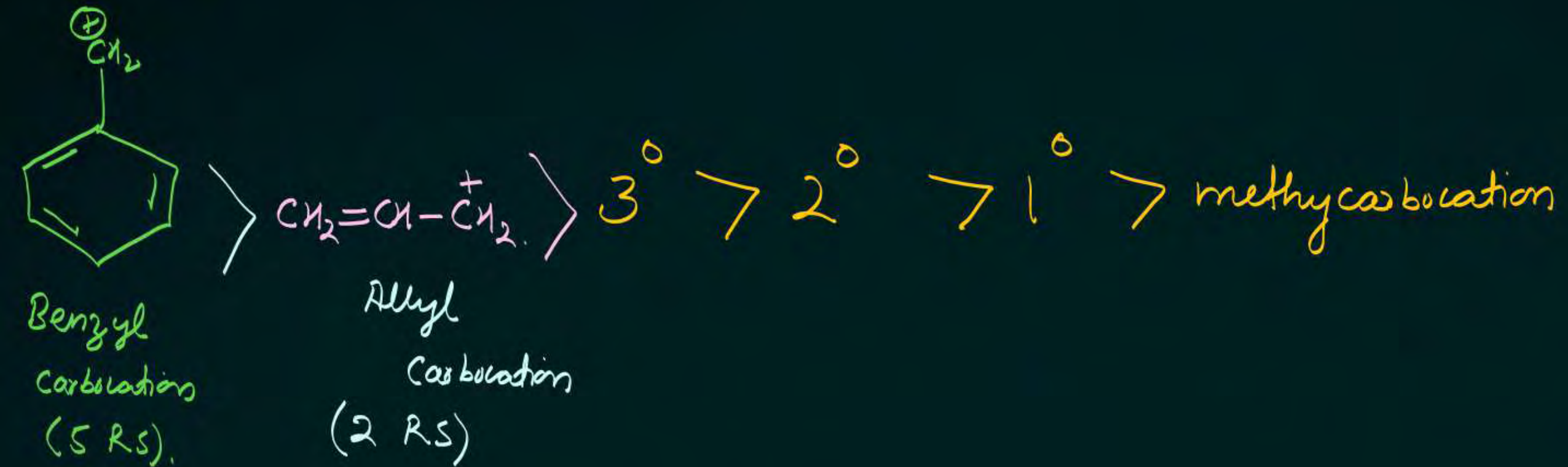


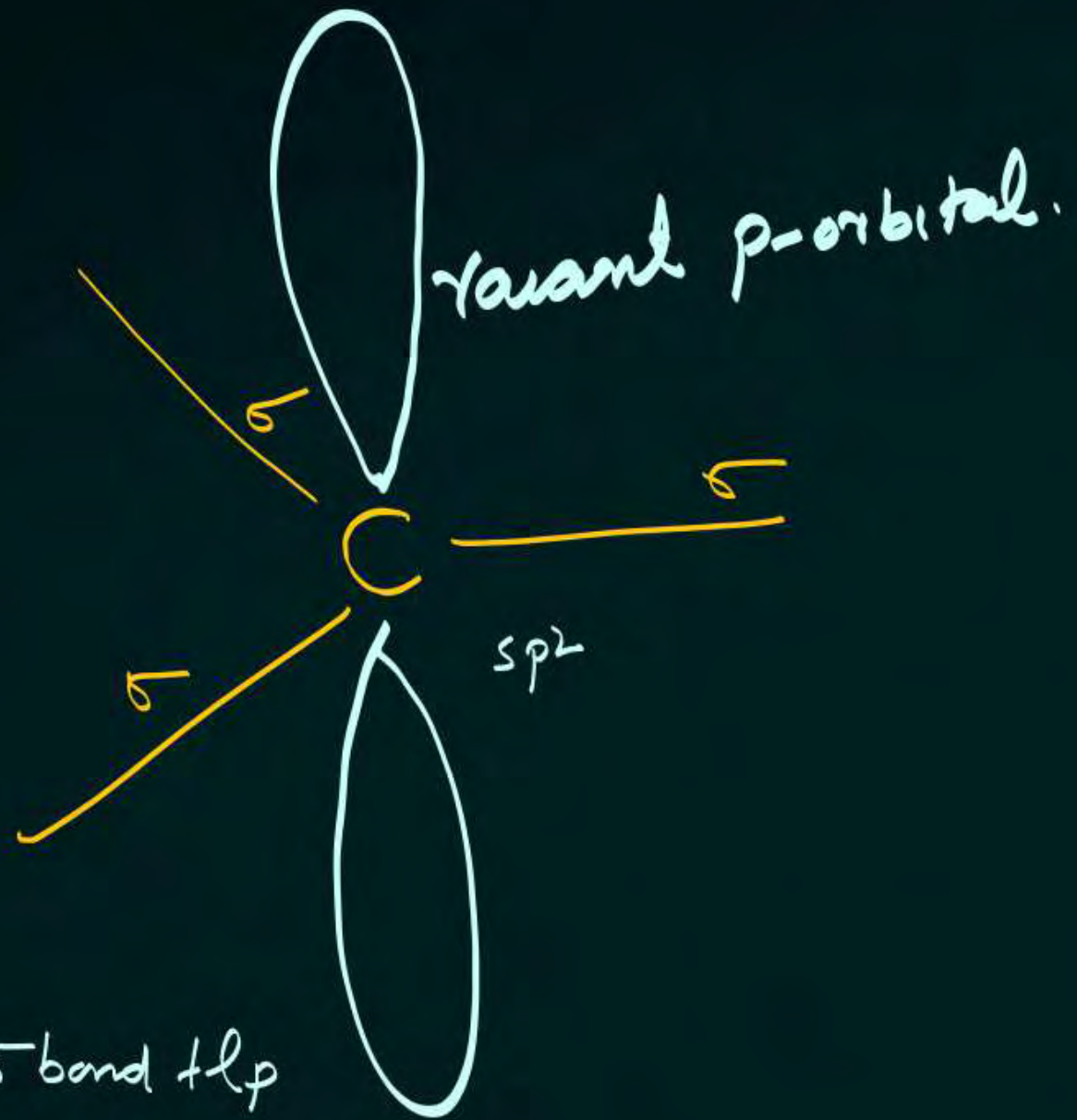
- C^+ is $\bar{e}$ deficient species $\rightarrow$ stabilised by EDG

3°C^+ three $\bar{e}$ donating group.

- Alkyl $\rightarrow$ +I
 $\rightarrow$ Hyperconjugation







$$\text{TEP} = \sigma \text{ bond} + \text{lp}$$

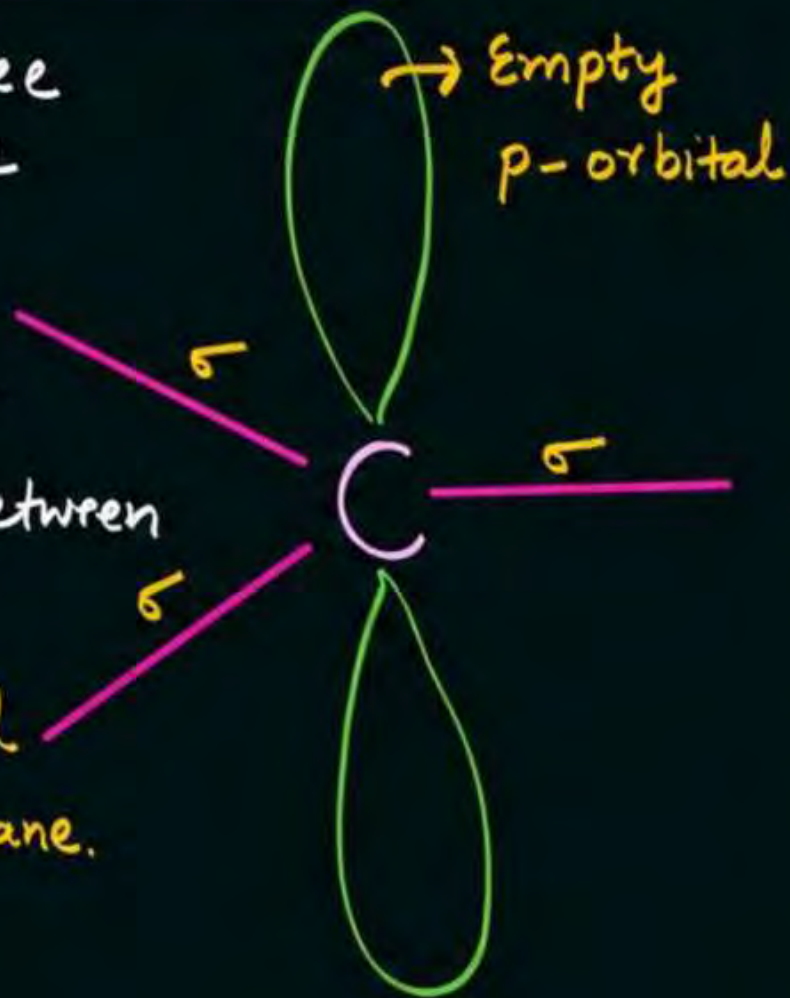
$$= 3 + 0$$

$$= 3$$

$$= sp^2$$

Orbital Structure of carbocation:→

- Carbon atom has three sigma bond & hence sp^2 hybridised
- It is trigonal planar with bond angle between σ -bonds is 120°
- One vacant-p-orbital is present $\perp$ to the plane.



$$TEP = \sigma + lp$$

2. Carbanion:→

Chemical Species bearing a negative charge on carbon and having eight e^- 's in their valence shell are called carbanion.

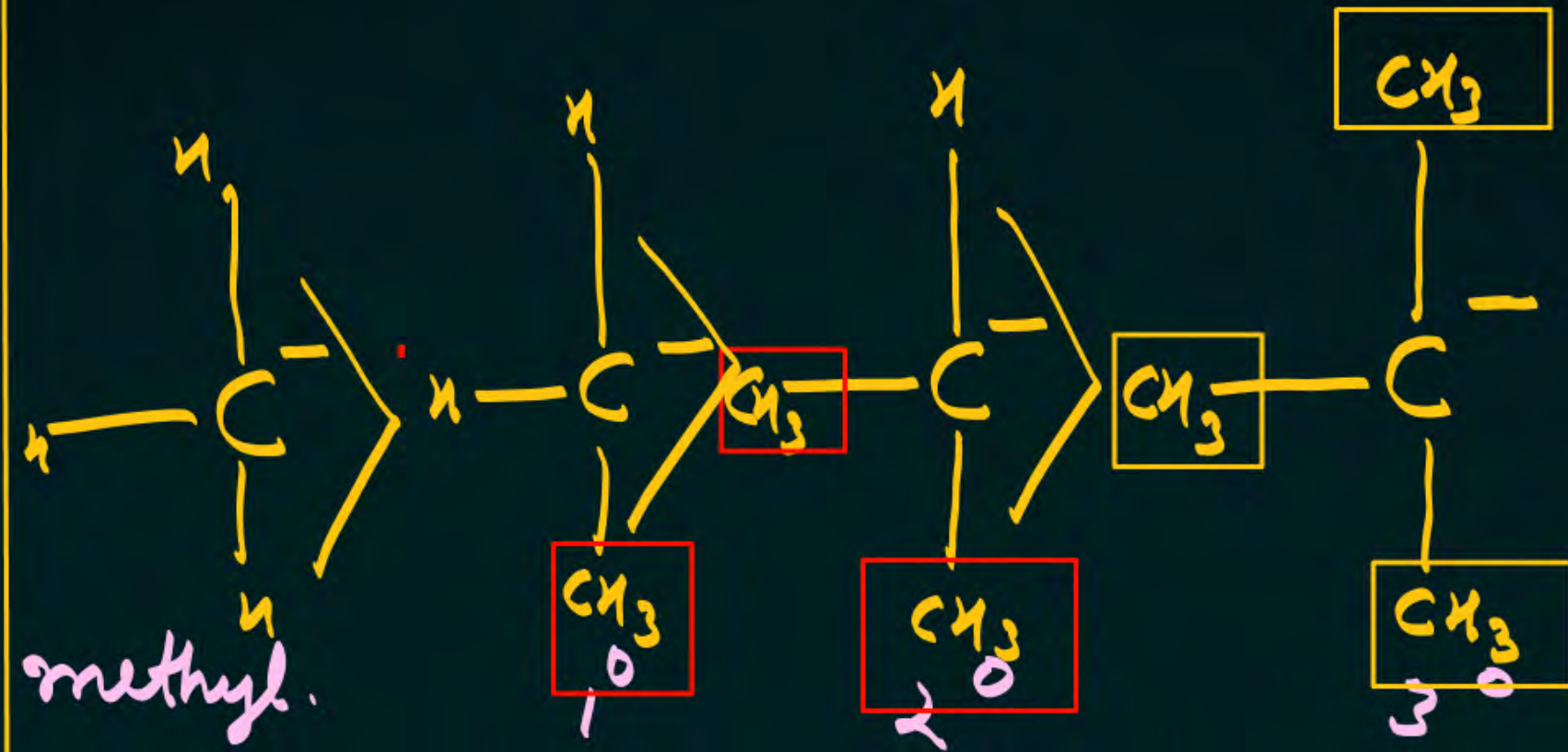
- They are produced by heterolytic cleavage.



It is e^- rich species.



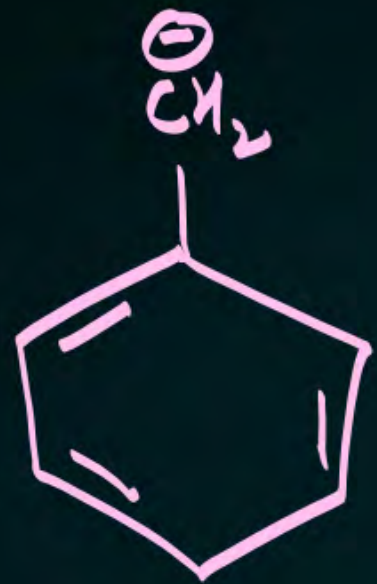
2) Carbanion $\rightarrow$



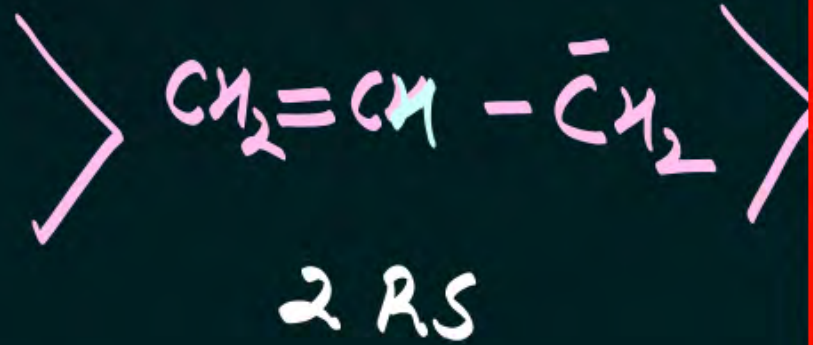
$\text{C}^{\ominus} \rightarrow$ stable EWG

$\rightarrow$ Unstable EDG. (3° $\bar{C}$ three $\bar{E}$ donating Group)

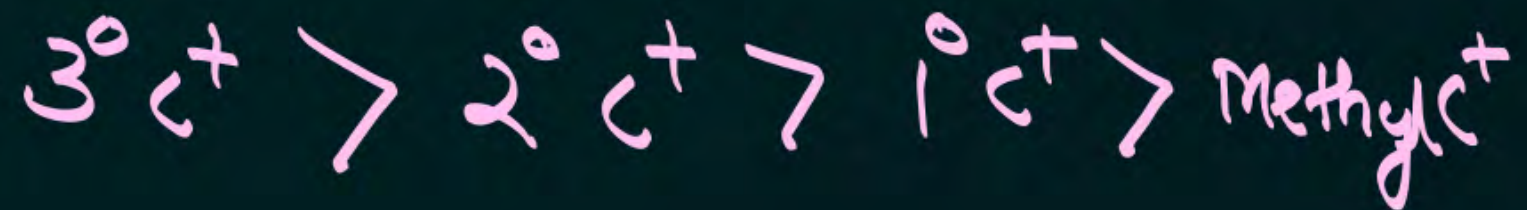
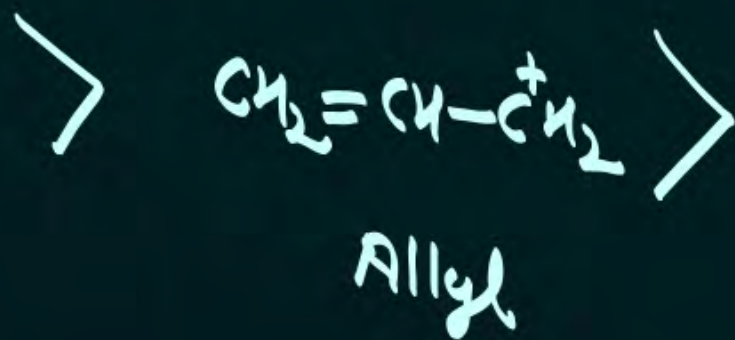
Allyl carbanion



5 RS

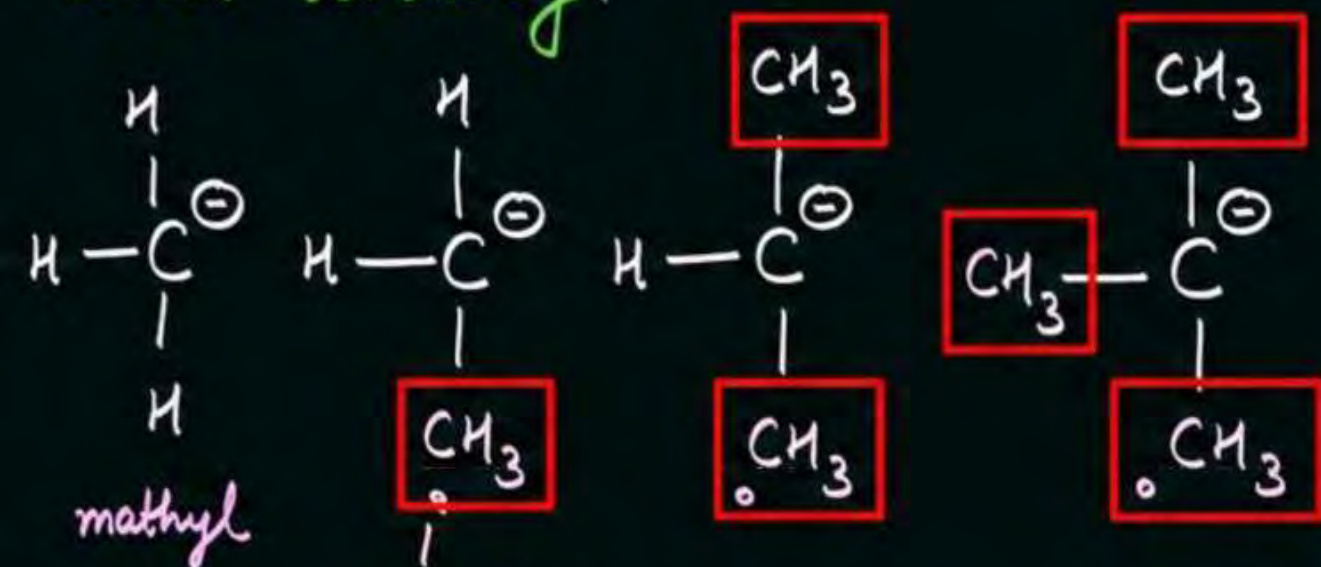


methyl
Carbanion



Classification of carbanion: →

It is also of 3 types i.e. primary, secondary and tertiary.



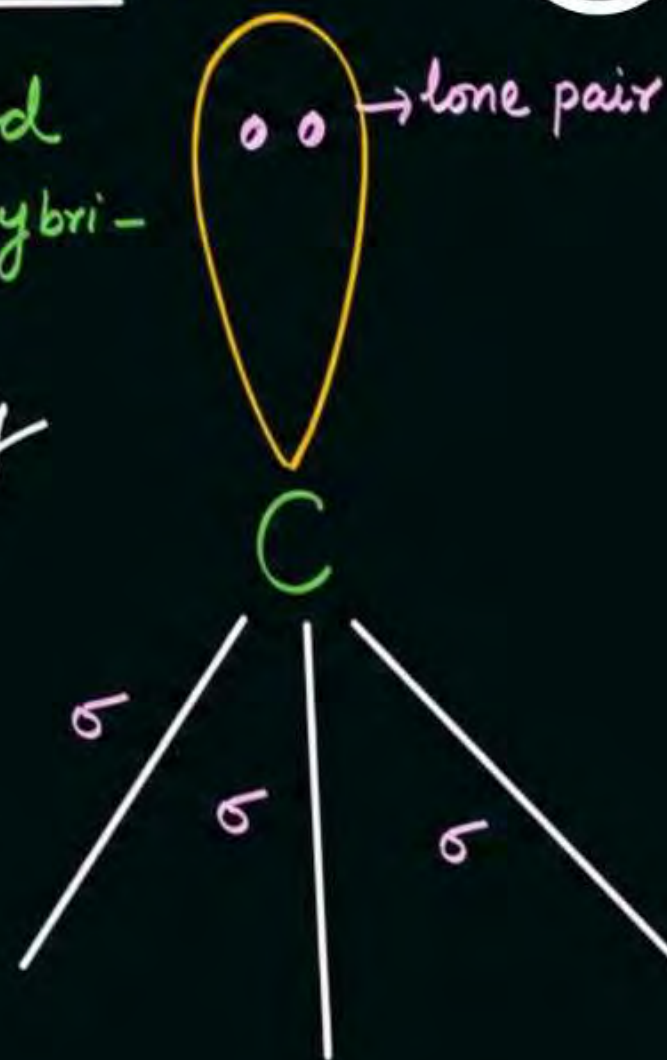
Order of Stability: → is methyl $> 1^\circ > 2^\circ > 3^\circ$

Because carbanion is $\bar{e}$ -rich species, therefore $\bar{e}$ withdrawing group will stabilise it. & $\bar{e}$ donating group will destabilise it.

3° carbanion has three $\bar{e}$ donating groups (via Inductive) therefore is least stable.

Orbital structure of Carbanion: →

- Carbon has 3- σ bonds and one lone pair, hence sp^3 hybridised.
- It has tetrahedral geometry and pyramidal shape.
- lone-pair is present in sp^3 hybridised orbital.



$$\begin{aligned} \text{TEP} &= \sigma + \text{lp} \\ &= 3 + 1 \\ &= 4 \rightarrow sp^3 \end{aligned}$$

Hyb = sp^3
 Geometry = Tetrahedral
 Shape = Pyramidal

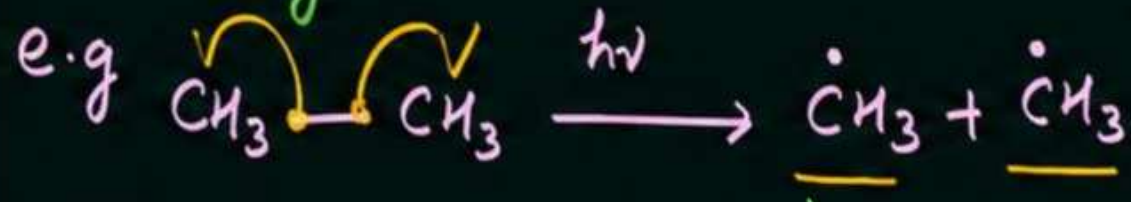


$$\begin{aligned} \text{TEP} &= \sigma + \text{lp} \\ &= 3 + 1 \\ &= 4 \\ &= sp^3 \end{aligned}$$

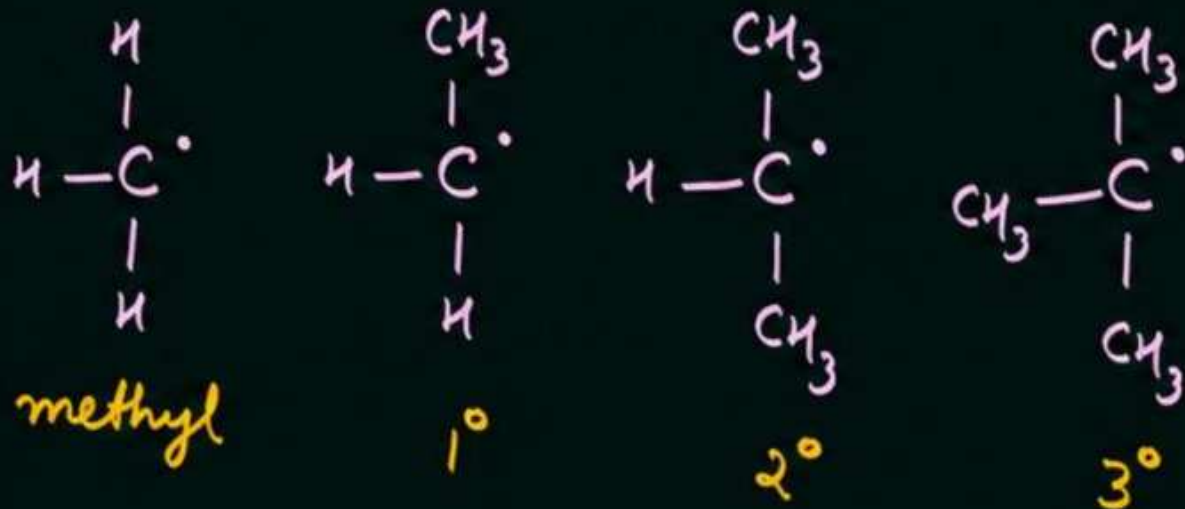
3) Free-Radicals $\Rightarrow$ Species having odd $\bar{e}$ in their valence shell are called free-radical.

- Carbon free radical is a neutral species having 7 $\bar{e}$ in their valence shell.

It is produced by homolytic cleavage.

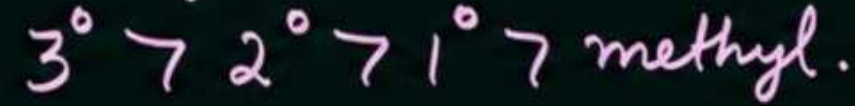


Classification $\Rightarrow$ Depending upon the no. of carbons attached, it is of 3 types, 1° , 2° & 3°



Stability $\Rightarrow$

Stability of free Radicals follows the order



As carbon free Radical is $\bar{e}$ deficient species, $\bar{e}$ donating group will stabilise it.

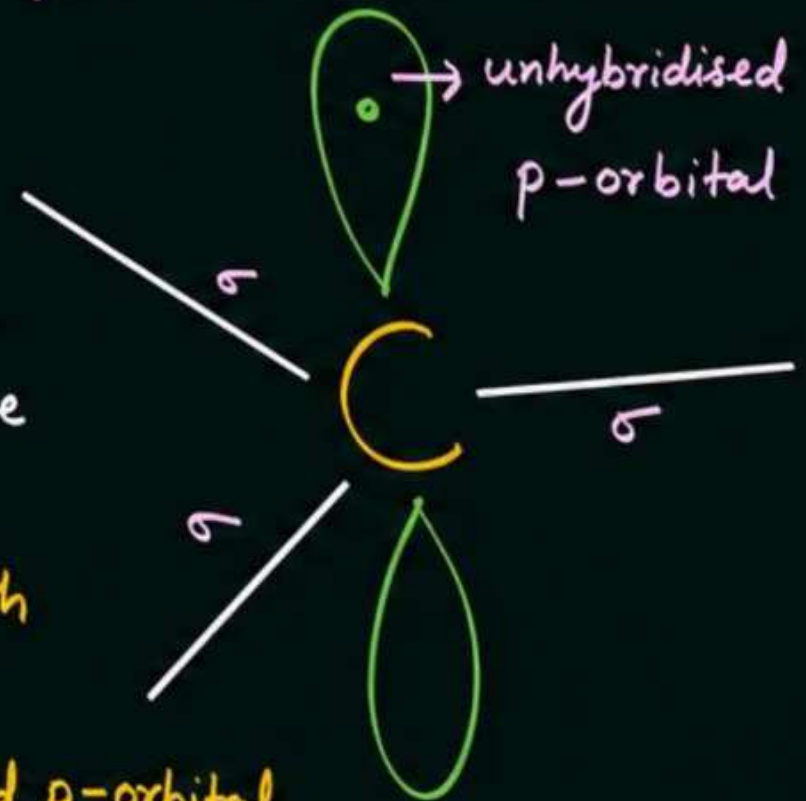
Tertiary free radical has three $\bar{e}$ donating groups attached, therefore has maximum stability.

Orbital Structure $\Rightarrow$

- As Carbon has 3 σ -bonds and one odd $\bar{e}$, therefore sp^2 hybridised.

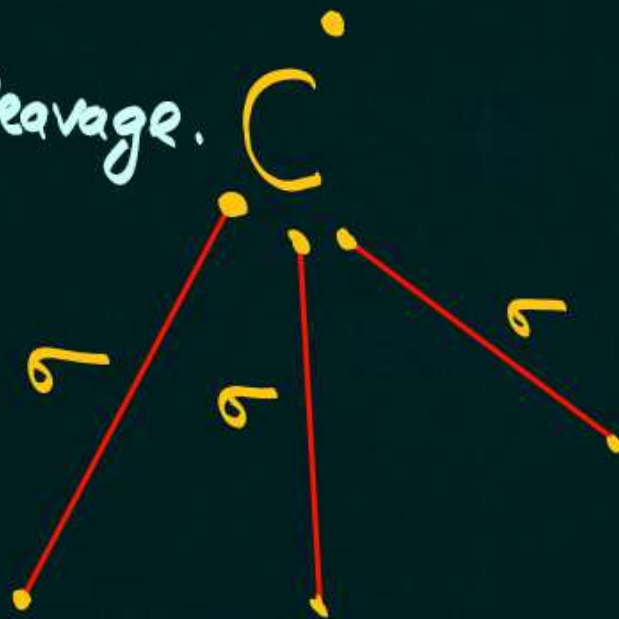
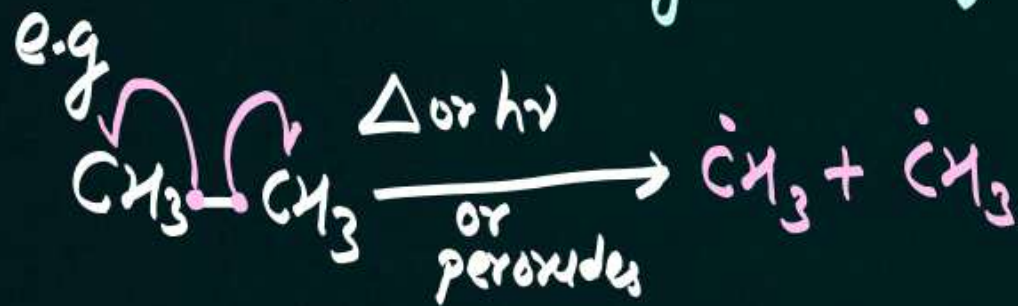
- Trigonal planar shape with bond angle of 120°

- Odd $\bar{e}$ is present in unbridised p-orbital present $\perp$ to the plane.

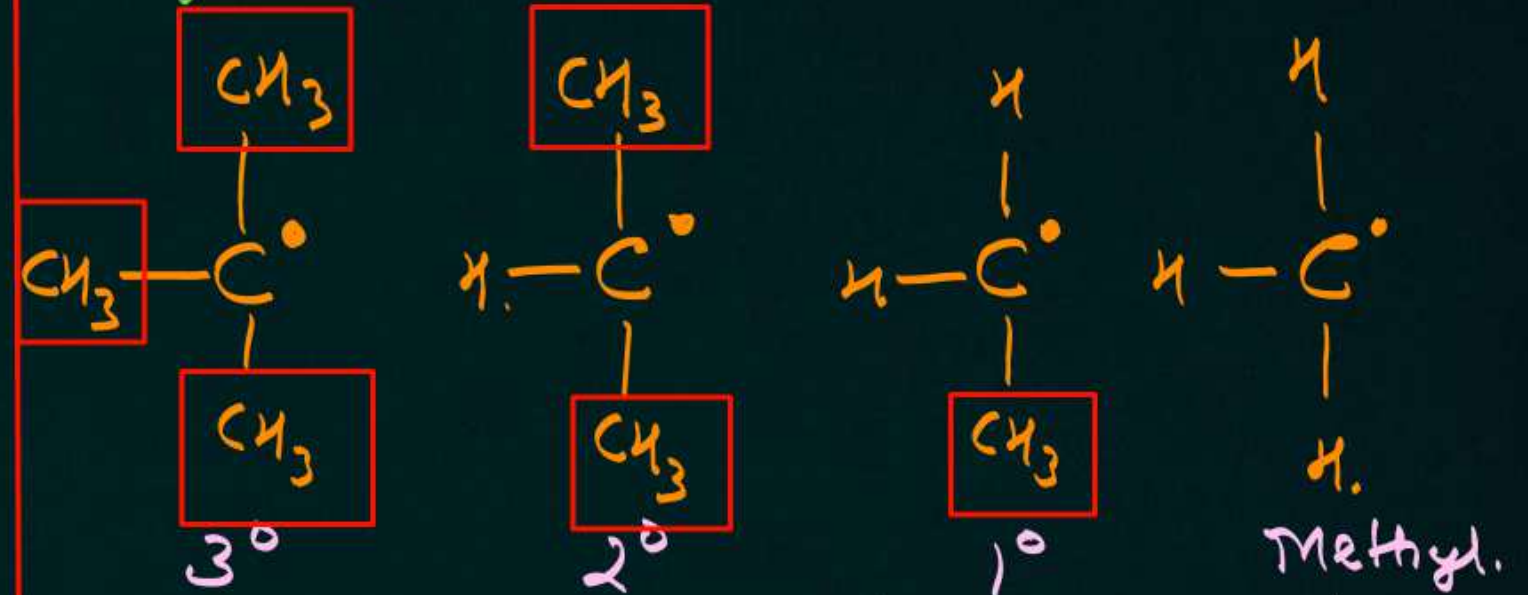


3) FREE RADICAL $\rightarrow$ Species having odd $\bar{e}$ in their valence shell is called free radical.

- Carbon free radical is defined as neutral species having 7 $\bar{e}$ in their valence shell.
- It is $\bar{e}$ deficient species.
- It is formed by homolytic cleavage.



Classification Depending upon the no. of carbon attached, Carbon free radical is of 3 types. 1° , 2° & 3° .



- As we know free radical is $\bar{e}$ deficient species therefore it is stabilised by any $\bar{e}$ donating group. 3° carbon free radical has three $\bar{e}$ donating groups hence is most stable. Order can be written as.

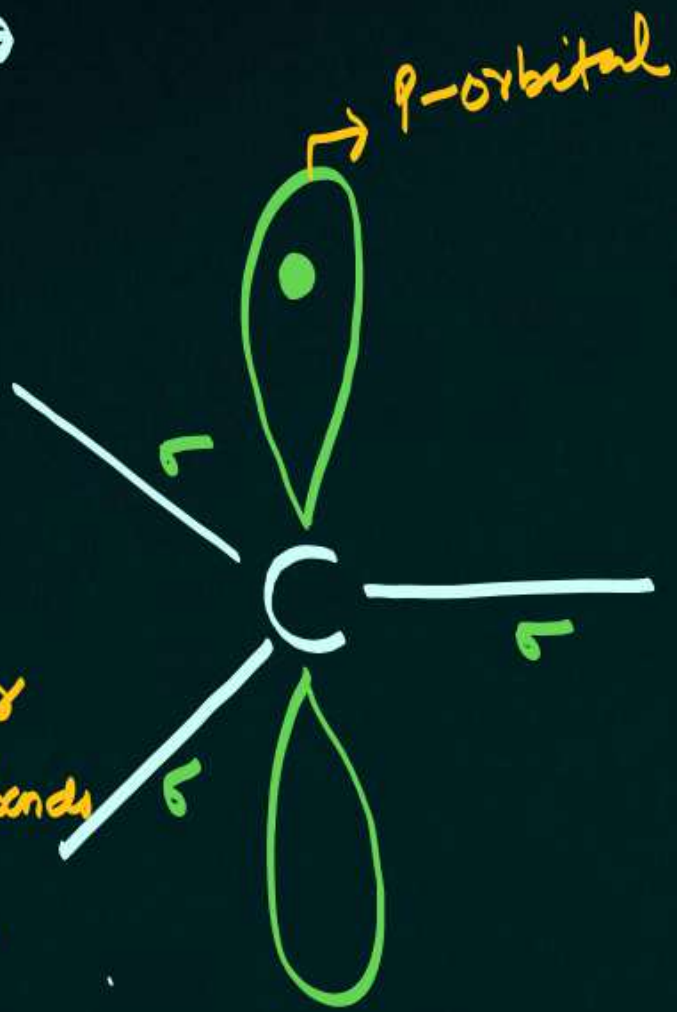


ORBITAL STRUCTURE →

As free radical has 3 sigma bonds hence it is sp^2 hybridised.

We can say that it is trigonal planar with all the three bonds are present in one plane.

P-orbital containing odd e^- is present perpendicular to the plane.



$$\begin{aligned} \text{TEP} &= \sigma + 2p \\ &= 3 + 0 \\ &= 3 \\ \text{i.e. } sp^2 \end{aligned}$$

TYPES OF ORGANIC REACTIONS.

There are four types of organic reactions:→

a) SUBSTITUTION REACTION

b) ADDITION RXN.

c) ELIMINATION Reaction

d) Rearrangement reaction.

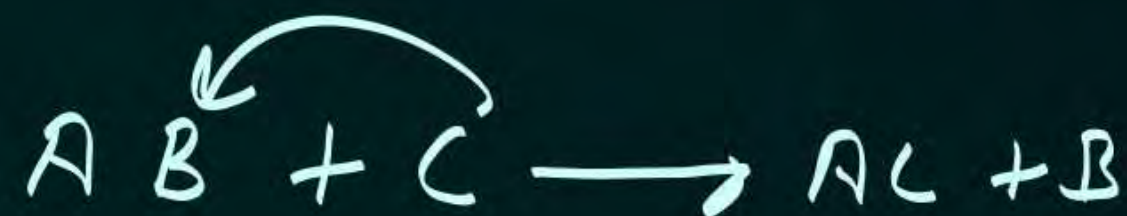
MOVIE

Dashrat
Manghi

Types of Reactions

Four Types of Organic Reactions

- (a) Substitution
- (b) Addition RxN.
- (c) Elimination RxN.
- (d) Rearrangement



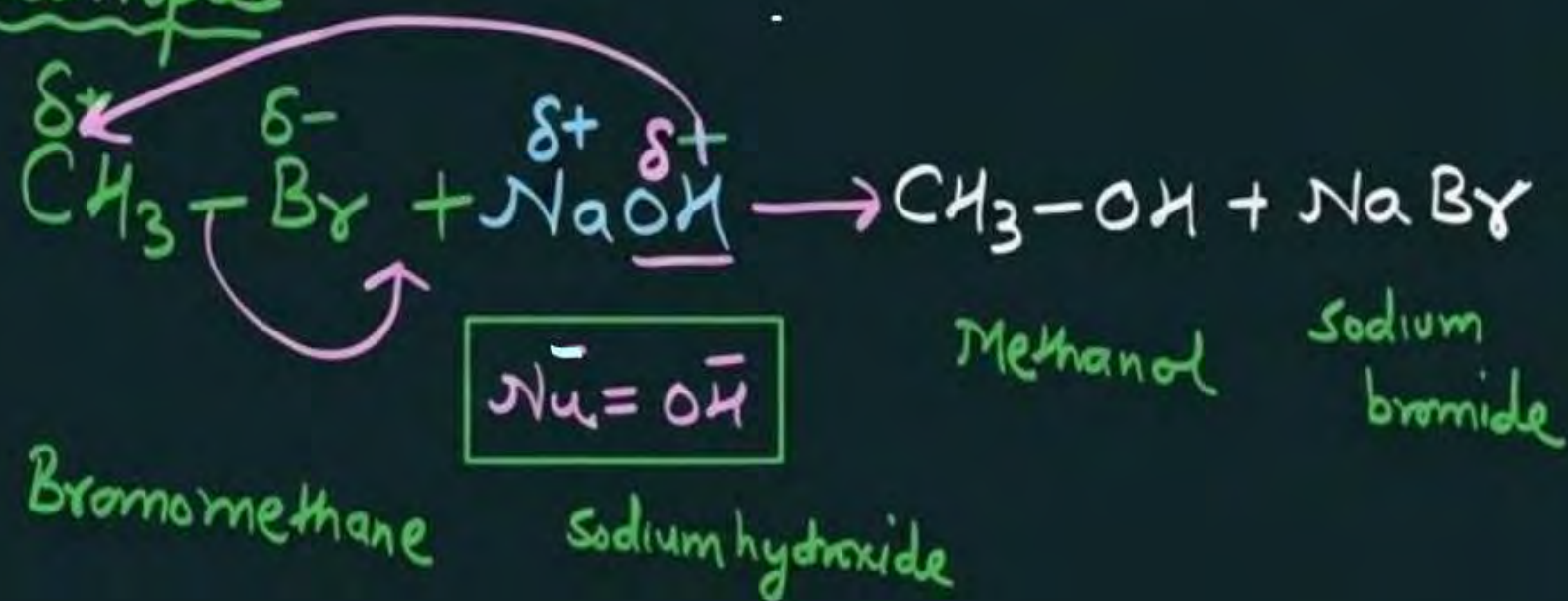
1. Substitution RxN

The reactions in which an atom or group of atoms in a molecule is replaced or substituted by different atoms or group of atoms are called substitution reactions.

① Nucleophilic Substitution Rxn.

Type of a substitution Rxn, which is initiated by nucleophile is called nucleophilic substitution Rxn.

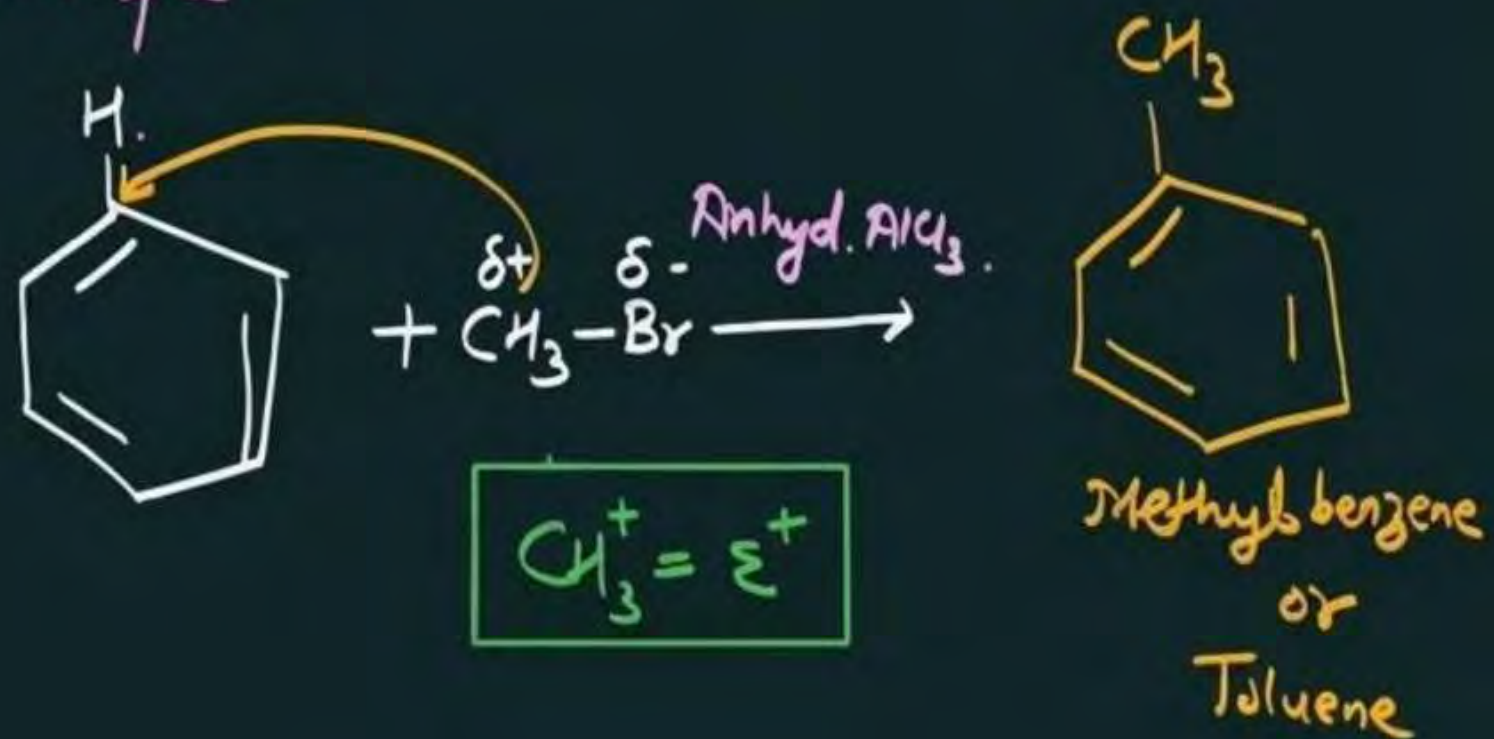
Example



2) Electrophilic Substitution Rxn.

Type of a substitution Rxn, which is initiated by electrophile is called electrophilic substitution Rxn.

Example

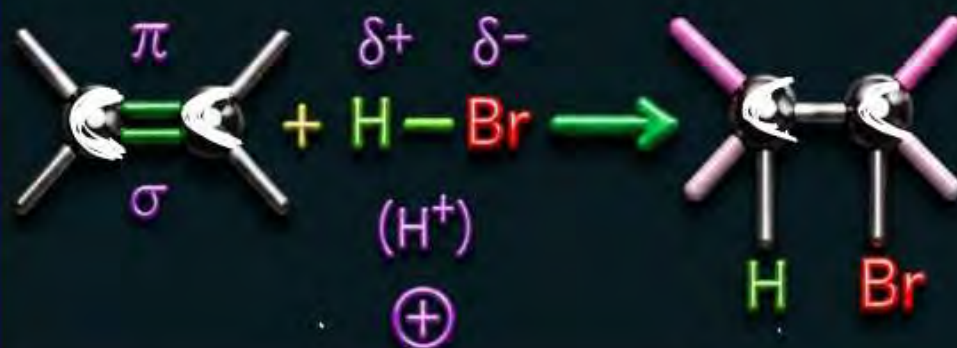


Types of Reactions

2. Addition reactions *Type of organic reaction in which two or more reactants combine to form one product.*

a) ELECTROPHILIC ADDITION RXN.

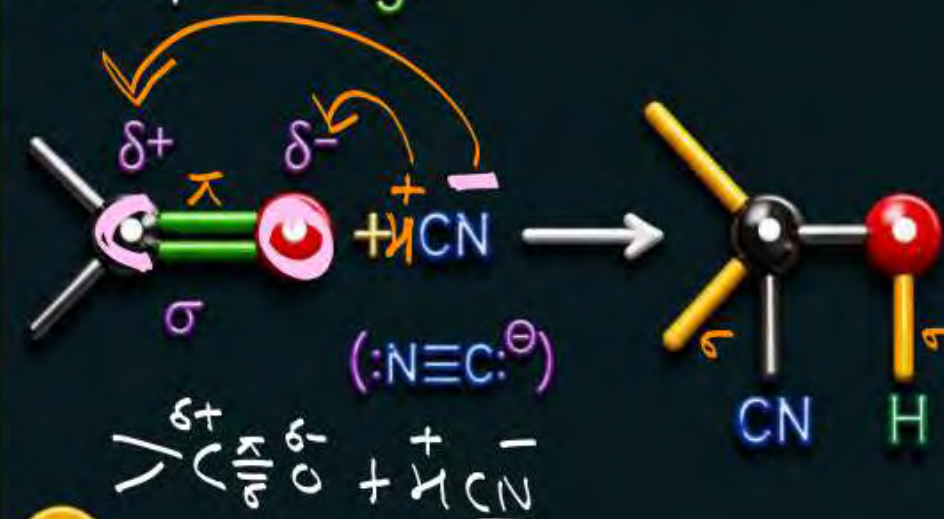
Type of addition Rxn which is initiated by electrophile. e.g.



- 1 π bond break and
- 2 σ -bond formation takes place.

b) NUCLEOPHILIC ADDITION RXN.

Type of addition Rxn, which is initiated by nucleophile. e.g.



- 1 π -bond break
- 2 σ -bond formation

CHALLENGE ACCEPTED.

Thank
You

#XUV
❤️